The Diaspora App - Android Google Sign-In Fix

Prepared for NDN Analytics | Current issue: Google Sign-In works on laptops/PCs but fails in the Android Play testing app with error 10.

> **Short answer:** the app code and web Google sign-in are working. Android error `10` means Google does not recognize the signing certificate of the Android app installed from Play. Firebase currently has the upload-key SHA, but Play installs are signed by Google Play App Signing with a different certificate.

1. What Is Happening

Platform	Google Auth Path	Status
Laptops and PCs	Firebase Web Auth using the web OAuth client on `www.thediaspora.app`.	Works.
Android Play testing app	Native Android Google Sign-In. Google checks package name plus Android signing certificate.	Fails with error `10`.

Error `10` is commonly Google Sign-In `DEVELOPER_ERROR`. In this case, the package name is correct, so the missing piece is the Google Play App Signing certificate fingerprint in Firebase.

2. Current Known App Values

Item	Value
Firebase project	`the-diaspora-app`
Firebase Android app ID	`1:530318956241:android:25ad2e43002e43376717f0`
Android package name	`com.ndnanalytics.thediaspora`
Live site	`https://www.thediaspora.app`
Latest Play AAB built	`builds/the-diaspora-app-native-v10-google-fix.aab`
Latest version	`1.0.3`, version code `10`

3. What Is Already Registered In Firebase

Firebase currently has the upload-key certificate:

```
SHA-1:
7E:1D:C1:DF:D6:0E:7C:10:FC:C8:2B:0B:0D:C3:AB:FE:B6:87:EC:6E

SHA-256:
09:BD:2B:53:0E:BF:C8:52:39:F2:F4:8B:E5:BA:72:8B:4D:74:73:AC:49:24:A0:7A:5A:01:4F:D4:39:DC:6A:7C
```

Important: the upload key is not the same certificate used when testers install from Google Play. Google Play re-signs the app with the Play App Signing key. That Play signing SHA-1 and SHA-256 must also be added to Firebase.

4. Permanent Fix

1. Get the **App signing key certificate** fingerprints from Google Play Console.
2. Add the Play App Signing **SHA-1** and **SHA-256** to the Firebase Android app.
3. Have testers fully close and reopen the Play-installed app, then try Gmail again.

Firebase CLI commands after the Play signing SHA values are known

```
firebase apps:android:sha:create 1:530318956241:android:25ad2e43002e43376717f0 PLAY_SIGNING_SHA1 --project the-diaspora-app

firebase apps:android:sha:create 1:530318956241:android:25ad2e43002e43376717f0 PLAY_SIGNING_SHA256 --project the-diaspora-app
```

In most cases, no new AAB is needed after adding these Firebase fingerprints because Google validates the app identity server-side against the Google/Firebase project.

5. How To Get The Missing Play Signing SHA

Option A: Play Console

1. Open Google Play Console.
2. Select The Diaspora App.
3. Go to **Test and release** > **Setup** > **App signing**.
4. Find **App signing key certificate**.
5. Copy **SHA-1 certificate fingerprint** and **SHA-256 certificate fingerprint**.

Option B: Extract from a tester phone

1. Install the app from the Play internal testing link on a tester phone.
2. Enable Developer options and USB debugging on the phone.
3. Connect the phone to this computer and approve USB debugging.
4. Run ADB to pull the Play-installed APK, then extract its signing certificate.

This method bypasses the Play Console UI and directly reads the certificate from the app as installed by Google Play.

6. Immediate Tester Workaround

A signed release APK was built with the upload key that Firebase already trusts:
`builds/the-diaspora-app-native-v10-google-fix-upload-key.apk`

To use it, testers must uninstall the Play-installed version first, then install this APK manually. Gmail should work in that sideloaded build because it is signed with the upload key already registered in Firebase.

7. Testing Checklist

- After Play App Signing SHA is added to Firebase, use the Play-installed app first.
- Force close the app, reopen it, and tap Gmail sign-in.
- If Gmail succeeds, verify the tester reaches `feed.html`.
- If Gmail still fails, capture the exact displayed error text and the tester device model.
- Email/password login remains available and should be used to keep testing moving.

8. Bottom Line

The Android app is failing because Google Play signs the app with a certificate that Firebase does not yet know. Add the Play App Signing SHA-1 and SHA-256 to Firebase, then the Play testing app should be able to use Gmail sign-in like the desktop web app already does.